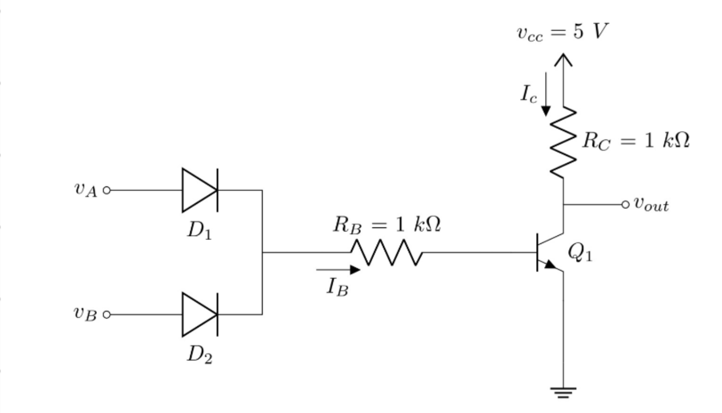


Set - A Question - 1

(a) Q_1 saturation $\Rightarrow V_{BE} = 0.8V$

$V_D = 0.7V$

$V_{CE} = 0.1V$

thus, $I_B = \dfrac{5-0.7-0.8}{1k} = 3.5mA$ 2

$I_C = \dfrac{5-0.1}{1k} = 4.9mA$ 2

(b) When $V_A = V_B = 5V$,

D_1 - forward bias 1

D_2 - forward bias 1

Q_1 - Saturation 1

(c) Power $= 5 \times (3.5 + 4.9) = 42$ mW 4

(d) When $V_A = V_B = 0.1V$

D_1 - reverse bias 1

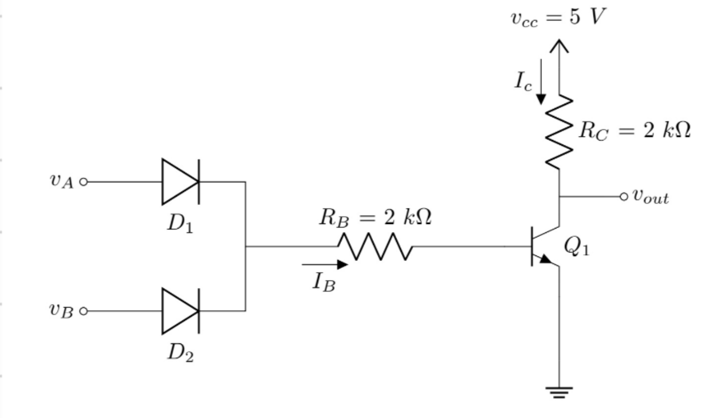
D_2 - reverse bias 1

Q_1 - cutoff 1

(e) NOR Gate 2

A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

4



Set - B Question - 4

(a) Q_1 saturation $\Rightarrow V_{BE} = 0.8V$

$V_D = 0.7V$

$V_{CE} = 0.1V$

thus, $I_B = \dfrac{5-0.7-0.8}{2k} = 1.75mA$ 2

$I_C = \dfrac{5-0.1}{2k} = 2.45mA$ 2

(b) When $V_A = V_B = 5V$,

D_1 - forward bias 1

D_2 - forward bias 1

Q_1 - Saturation 1

(c) Power $= 5 \times (1.75 + 2.45) = 21$ mW 4

(d) When $V_A = V_B = 0.1V$

D_1 - reverse bias 1

D_2 - reverse bias 1

Q_1 - cutoff 1

(e) NOR Gate 2

A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

4

Set-A Q(2)

(a)

D_A just ON, T_1 D_2 T_2 ON

$\therefore V_P = V_A + V_N + 0.6 = 10.6 + V_N$

again, $V_P = V_{BE2} + V_{D2} + V_{BE1} = 0.8 + 0.7 + 0.7 = 2.2$

$\therefore V_N + 10.6 = 2.2 \rightarrow V_N = -8.4$

$\hookrightarrow |V_N| = 8.4$ $\boxed{\text{Ans}}$

(b)

$$\frac{10 - V_{c1}}{14k\Omega} = \left(\frac{V_c - 2.2}{15k\Omega}\right) \times 26$$

$\hookrightarrow V_c = 2.509V$

$I_1 = \frac{10 - 2.509}{14k\Omega} = 0.535mA$ $\boxed{\text{Ans}}$

$I_{B2} = I_1 - \frac{0.8}{90k\Omega} = 0.526mA$ $\boxed{\text{Ans}}$

$I_3 = \frac{10 - 0.1}{2k\Omega} = 4.95mA$ $\boxed{\text{Ans.}}$

Load $\longrightarrow I_1 = \frac{10 - 0.8}{29k\Omega} = 0.317\boxed{mA}$ $\boxed{\text{Ans.}}$

(c)

$$\frac{I_3 + 90 \times 0.317}{0.526} < 25$$

$\hookrightarrow I_3 = 6.81mA \longrightarrow \frac{10 - 0.1}{R_c} = 6.81mA \rightarrow \boxed{R_c = 1.45k\Omega}$

OR, $\boxed{1.454k\Omega < R_c < 1.525k\Omega}$

Single value or Range both valid.

Set-B (Q3)

(a)

D_A just ON, T_1 D_2 T_2 ON

$\therefore V_P = V_A + V_N + 0.6 = 10.6 + V_N$

again, $V_P = V_{BE2} + V_{D2} + V_{BE1} = 0.8 + 0.7 + 0.7 = 2.2$

$\therefore V_N + 10.6 = 2.2 \rightarrow V_N = -8.4 \rightarrow |V_N| = 8.4V$

$\boxed{\text{Ans}}$

(b)

$$\frac{10 - V_{c1}}{13k\Omega} = 36 \times \left(\frac{V_{c1} - 2.2}{16k\Omega}\right)$$

$\hookrightarrow V_{c1} = 2.458V$

$I_1 = \frac{10 - 2.458}{13k\Omega} = 0.58mA$ $\boxed{\text{Ans}}$

$I_{B2} = I_1 - \frac{0.8}{80k\Omega} = 0.57mA$ $\boxed{\text{Ans}}$

$I_3 = \frac{10 - 0.1}{2.4k\Omega} = 4.125mA$ $\boxed{\text{Ans}}$

Load $\rightarrow I_1 = \frac{10 - 0.8}{29k\Omega} = 0.317mA$ $\boxed{\text{Ans}}$

(c)

$$\frac{I_3 + 40 \times 0.317}{0.57} < 35$$

$\hookrightarrow I_3 = 7.27mA \rightarrow \frac{10 - 0.1}{R_c} = 7.27mA \rightarrow \boxed{R_c = 1.362k\Omega}$

OR, $\boxed{1.362k\Omega < R_c < 1.424k\Omega}$

Single value or Range both valid.

Midterm – Set A

Q3

(a)

both inputs are low = 0.1V

$Q_1 \rightarrow$ saturation, $Q_2 \rightarrow$ cutoff, $Q_3 \rightarrow$ cutoff

$Q_4 \rightarrow$ Forward active

individual demand, $i'_L = 0.4 \times \frac{5-2.3}{6k\Omega}$

$= 0.18$ mA

total demand, $i_{E4} = \cancel{8 \times 0.18}$ 8×0.18

$= 1.44$ mA

2

So, total supply = 1.44 mA

Now, $I_{B4} = \frac{1.44}{(\beta+1)} = \frac{1.44}{41} = 0.0351$ mA → 2

Now,

$I_{B4} = \frac{5 - V_{B4}}{2k\Omega}$

$\cancel{V_{B4} = 4.93}$ $\boxed{V_{B4} = 4.93V}$ — 2

$\boxed{V_0 = 3.53V}$ ← 2

(b)

$I_{B4} = 0.0351$ mA

$I_{C4} = 40 \times 0.0351 = 1.404$ mA (2)

So, $I_{C4} = \dfrac{5 - V_{C4}}{20\,\Omega}$, V_{C4} $\cancel{=}$ 4.97192V.

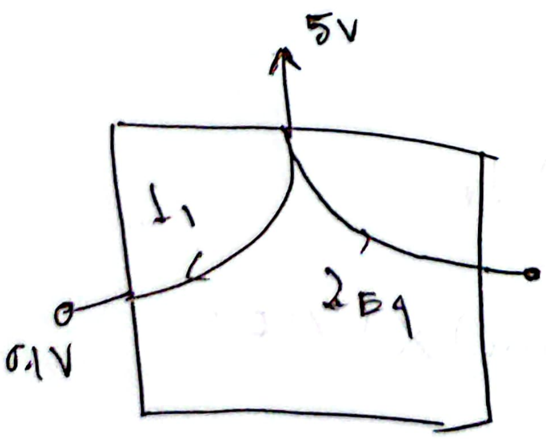
So $V_{C4} = 4.97192$ V (2)

$V_{B4} = 4.23$ V.

So. $(V_{CE})_4 = 0.742 > 0.1$. (2)

So, Q_4 in forward active [verified]

(c)



$I_{E4} = 1.44$ mA

$I_1 = \dfrac{5 - 0.9}{6\,k\Omega}$ (2)

$= 0.683$ mA

$P = (5 - 0.1) \times I_1 + (5 - 3.53) \times I_{E4}$

$\boxed{P = 5.4634 \text{ mW}}$ (4)

Set B

Q2

a)

$$i_L = 0.3\times\frac{5-2.3}{5k} = 0.162\,mA, \quad i_{E4} = 1.458\,mA$$

$$I_{B4} = \frac{1.458}{31} = 0.047\,mA$$

$\therefore V_{B4} = 4.906\,V.$

$\boxed{V_0 = 3.506\,V}$

b)

$$I_{C4} = 30\times 0.047 = 1.41\,mA$$

$$I_{C4} = \frac{5-V_{C4}}{40}, \quad V_{C4} = 4.9436\,V.$$

$V_{E4} = 4.206. \quad V_{CE4} = 0.737679$

so in active mode

c)

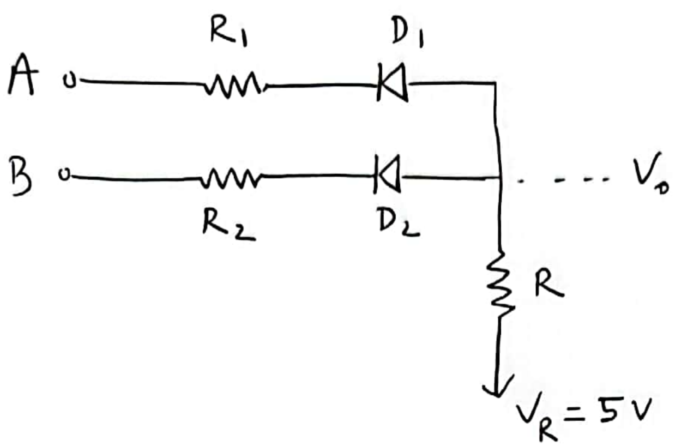
$$P = (5-0.1)\times I_1 + (5-V_0)\,I_{E4}$$

$$= (-0.1)\times 0.92 + 5(-3.506)\times 1.458$$

$\boxed{= 6.2\,mW}$

Set A

Question 4



$R_1 = R_2 = 680\,\Omega$

$R = 91\,K\Omega$

(a) $A = B = 5\,V$

D_1, D_2 both OFF.

$\boxed{V_o = 5\,V}$ **<u>Set B : 5 V</u>**

(b) $A = 5V, B = 0\,V$

or $A = 0\,V, B = 5\,V$

$$\frac{(V_o - 0.7) - 0}{R_1} = \frac{V_R - V_o}{R}$$

$$\Rightarrow \frac{V_o - 0.7}{R_1} = \frac{5 - V_o}{R}$$

$\Rightarrow \boxed{V_o = 0.7319\,V}$ (upto 4 decimal place)

<u>Set B: 0.7252 V</u>

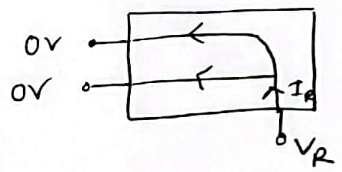
(c) $A = B = 0\,V$

$$2 \times \frac{V_o - 0.7}{R_1} = \frac{5 - V_o}{R}$$

$\Rightarrow \boxed{V_o = 0.716\,V}$ (upto 4 decimal place)

<u>Set B: 0.7126 V</u>

(b) Maximum power dissipation when $A = B = 0\,V$



$\therefore P = (V_R - 0) \times I_R$

$= (5 - 0) \times \frac{5 - V_o\,(0,0)}{R}$ **<u>Set B: 0.2257</u>**

$= 5 \times \frac{5 - 0.716}{91} = \boxed{0.2354\text{ mW}}$

(upto 4 d.p.)

(c) You can neglect and set $R_1 = R_2 = 0$ at the beginning of the calculation. Then, $V_o = 0.7\,V$

and, $P = 5 \times \frac{5 - 0.7}{R} < 1$

$\Rightarrow R > 21.5\,K\Omega$

$\therefore \boxed{R_{min} = 21.5\,k\Omega}$

<u>Set B: same</u>

Or, you can write 'P' as a function of 'R', 'R_1', 'R_2' and eliminate R_1, R_2 by comparing with R. This should result in the same value but with a lot of calculation.